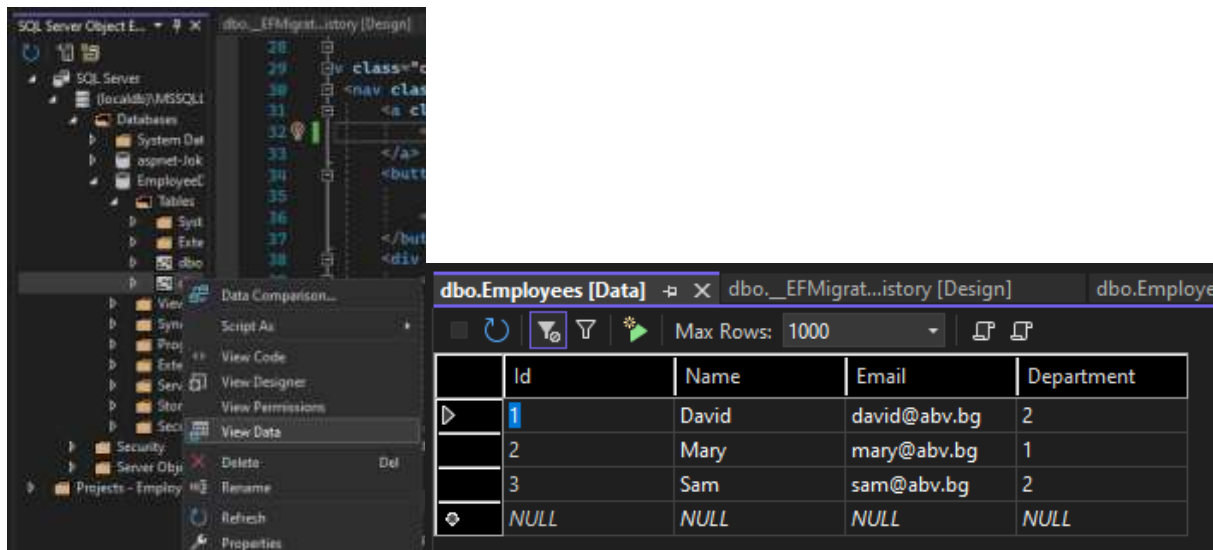


51 - Entity framework core seed data

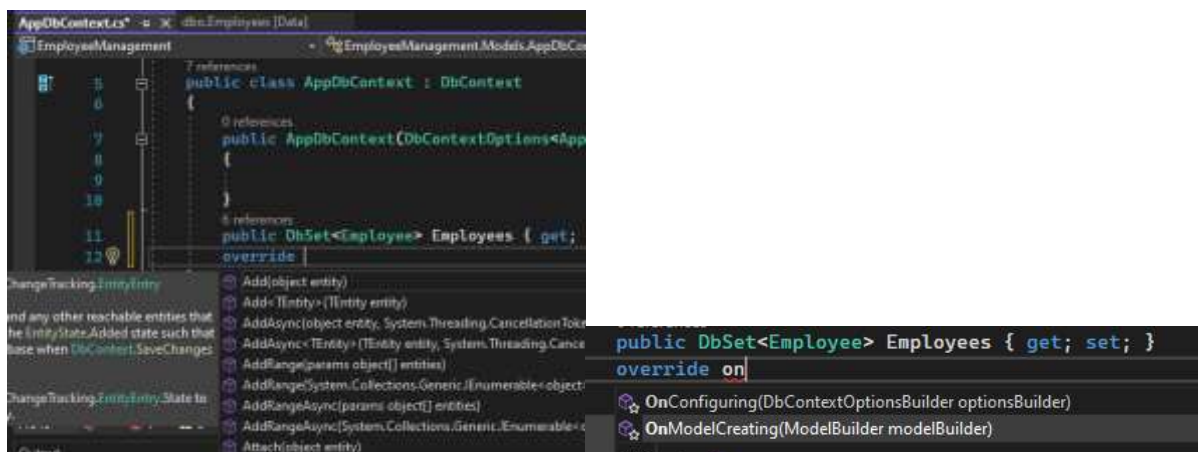
Table -> View data, въведох малко данни през сайта.



The screenshot shows the SQL Server Enterprise interface. On the left, the 'Server Object Explorer' displays the database structure. The main pane shows the 'dbo.Employees [Data]' table with 1000 rows. The first four rows are visible:

	Id	Name	Email	Department
▶	1	David	david@abv.bg	2
	2	Mary	mary@abv.bg	1
	3	Sam	sam@abv.bg	2
⊕	NULL	NULL	NULL	NULL

Другият начин е през AppDbContext, команда **override** -> **OnModelCreating**



The screenshot shows the 'AppDbContext' class in the 'EmployeesManagement.Models' namespace. The 'OnModelCreating' method is overridden, and the 'Employees' DbSet is configured.

```
public class AppDbContext : DbContext
{
    public AppDbContext(DbContextOptions<AppDbContext> options) : base(options)
    {
    }

    public DbSet<Employee> Employees { get; set; }

    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        // Add object entity
        // Add entity (TEntity entity)
        // AddAsync(object entity, System.Threading.CancellationToken)
        // AddAsync(TEntity entity, System.Threading.CancellationToken)
        // AddRange(params object[] entities)
        // AddRange(System.Collections.Generic.IEnumerable<object> entities)
        // AddRangeAsync(params object[] entities)
        // AddRangeAsync(System.Collections.Generic.IEnumerable<object> entities)
        // Attach(object entity)
    }
}
```

автокод:

cbzalhawysy;'mxvnm,./vbnm,./qs

```
protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    base.OnModelCreating(modelBuilder);
}

protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    base.OnModelCreating(modelBuilder);
}
```

В конкретния случай, команда **base.OnModelCreating(modelBuilder);** не ни върши

работа, затова я изтриваме.

```
protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    modelBuilder.Entity<Employee>().HasData(
        new Employee {
            Id = 1,
            Name = "Mark",
            Department = Dept.IT,
            Email = "mark@abv.bg"
        }
    );
}
```

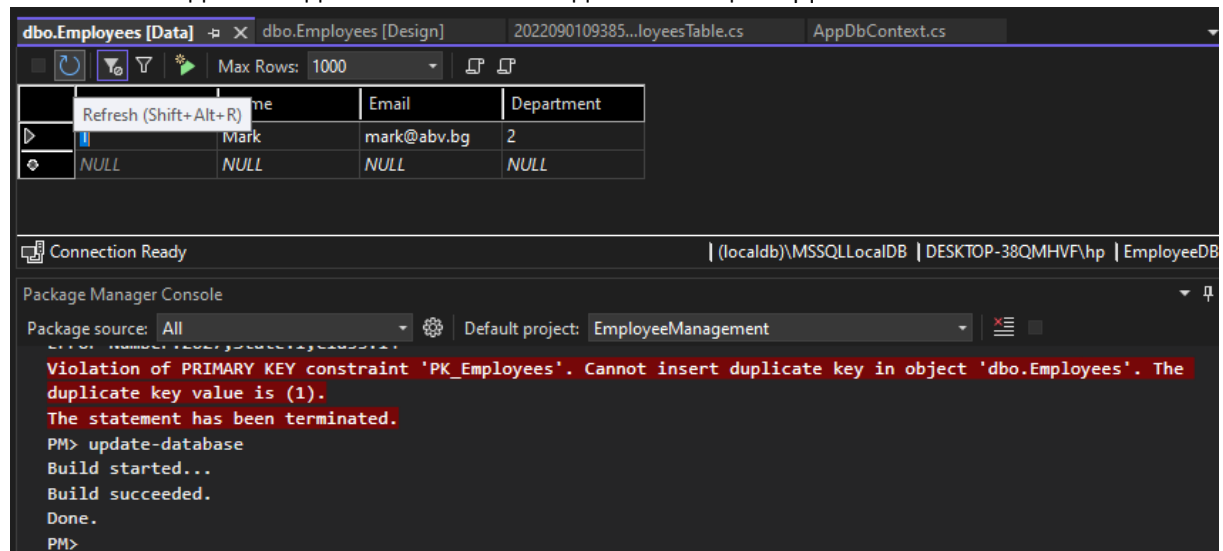
```
protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    modelBuilder.Entity<Employee>().HasData(
        new Employee {
            Id = 1,
            Name = "Mark",
            Department = Dept.IT,
            Email = "mark@abv.bg"
        }
    );
}
```

За да вмъкнем данните в БД, ни е необходима нова миграция:

View->Other windows -> Package Migrations ->

add-migration SeedEmployeesTable, update-database.

Нарочно въведох Id = 1, за да видя дали ще има презаписване, но при update изведе грешка, затова изтрих таблицата с данните и оставих само AppDbContext. Новият служител се въведе, което става видимо след най-важната команда на таблицата БД Refresh:



The screenshot shows the SQL Server Enterprise Designer interface. The top pane displays the 'dbo.Employees' table with columns: Id, Name, Email, and Department. The table contains one record with Id=2, Name=Mark, Email=mark@abv.bg, and Department=2. Below the table, the Package Manager Console shows the following error message:

```
Violation of PRIMARY KEY constraint 'PK_Employees'. Cannot insert duplicate key in object 'dbo.Employees'. The duplicate key value is (1).
The statement has been terminated.
PM> update-database
Build started...
Build succeeded.
Done.
PM>
```

това се отразява и в браузъра.

Ако сме допуснали грешка, въвели сме Mark, вместо Mary, тогава редактираме C-кода и правим нова миграция. HasData() позволява въвеждане на много служители.

```

modelBuilder.Entity<Employee>().HasData(
    new Employee {
        Id = 1,
        Name = "Mary",
        Department = Dept.IT,
        Email = "mary@abv.bg"
    },
    new Employee {
        Id = 2,
        Name = "John",
        Department = Dept.HR,
        Email = "john@abv.bg"
    }
);

```

	Refresh (Shift+Alt+R)	Name	Email	Department
1		Mary	mary@abv.bg	2
2		John	john@abv.bg	1
		NULL	NULL	NULL

Add-migration AlterEmployeeSeedData, получавам съобщение за презаписване на БД, update.

An operation was scaffolded that may result in the loss of data. Please review the migration for accuracy. To undo this action, use Remove-Migration.



Колкото повече hard code добавяме като C data -> AppDbContext, толкова по-разпилян ще е кодът.

```

AppDbContext.cs
EmployeeManagement.Models.AppDbContext
OnModelCreating(ModelBuilder modelBuilder)
{
    modelBuilder.Entity<Employee>().HasData(
        new Employee {
            Id = 1,
            Name = "Mary",
            Department = Dept.IT,
            Email = "mary@abv.bg"
        },
        new Employee {
            Id = 2,
            Name = "John",
            Department = Dept.HR,
            Email = "john@abv.bg"
        }
    );
}

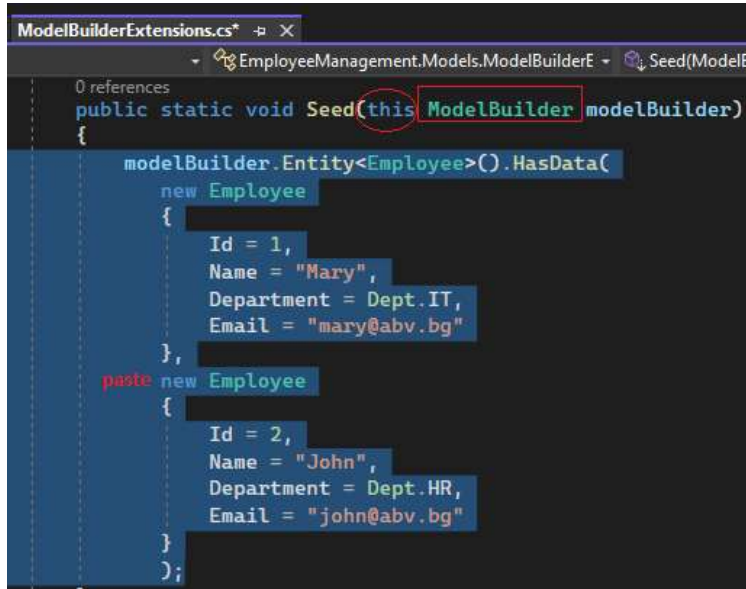
```

За да се структурира по-добре, може да се премести в допълнителен modelBuilder class.

Папка Models->Add-> Class->ModelBuilderExtensions

За Extensions е необходим static, освен това, за разширенията е необходимо да се посочи

откъде са, чрез ключова дума this и името на класа.



```
public static void Seed(this ModelBuilder modelBuilder)
{
    modelBuilder.Entity<Employee>().HasData(
        new Employee
        {
            Id = 1,
            Name = "Mary",
            Department = Dept.IT,
            Email = "mary@abv.bg"
        },
        new Employee
        {
            Id = 2,
            Name = "John",
            Department = Dept.HR,
            Email = "john@abv.bg"
        }
    );
}
```

На променливата modelBuilder присвояваме class Seed, който е разширение.

```
AppDbContext.cs
using Microsoft.EntityFrameworkCore;

namespace EmployeeManagement.Models
{
    9 references
    public class AppDbContext : DbContext
    {
        0 references
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options)
        {
        }
        6 references
        public DbSet<Employee> Employees { get; set; }
        0 references
        protected override void OnModelCreating(ModelBuilder modelBuilder)
        {
            modelBuilder.Seed();
        }
    }
}
```

```
using Microsoft.EntityFrameworkCore;
```

```
namespace EmployeeManagement.Models
{
```

```
    public class AppDbContext : DbContext
    {
```

```
        public AppDbContext(DbContextOptions<AppDbContext> options) :
base(options)
        {
```

```
        }
        public DbSet<Employee> Employees { get; set; }
```

```
        protected override void OnModelCreating(ModelBuilder modelBuilder)
        {
            modelBuilder.Seed();
        }
    }
```

```
}
```

Entity Framework Core Seed Data

```
public class AppDbContext : DbContext
{
    public AppDbContext(DbContextOptions<AppDbContext> options)
        : base(options)
    { }

    public DbSet<Employee> Employees { get; set; }

    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        modelBuilder.Entity<Employee>().HasData(
            new Employee
            {
                Id = 1, Name = "Mark", Department = Dept.IT,
                Email = "mark@pragimtech.com"
            }
        );
    }
}
```